# ROUND 1 REVIEW

## AI Frontier Challenge 2026 Round 1 Submission

**Team Name**

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**Team Leader**

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**Track**

H5. Frontline Health-Worker Companion for Low-Connectivity Areas

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# Use Case

## Problem Statement

- **Frontline Burden & Zero Cellular Connectivity:**
  Over 10.29 lakh rural ASHA & ANM workers navigate 1,700+ remote AP villages with zero mobile reception, juggling bulky flipbooks and 20+ disconnected apps.
- **Preventable Maternal & Neonatal Deaths:**
  Obstetric haemorrhage causes 47% of maternal deaths; missed newborn danger signs (hypothermia, convulsions, chest indrawing) lead to fatal referral delays.
- **Cloud AI Hallucinations & Malpractice Risks:**
  Generic cloud LLMs hallucinate fatal antibiotic dosages, violate non-prescribing ASHA boundaries, require constant 4G data, and fail on low-end smartphones.

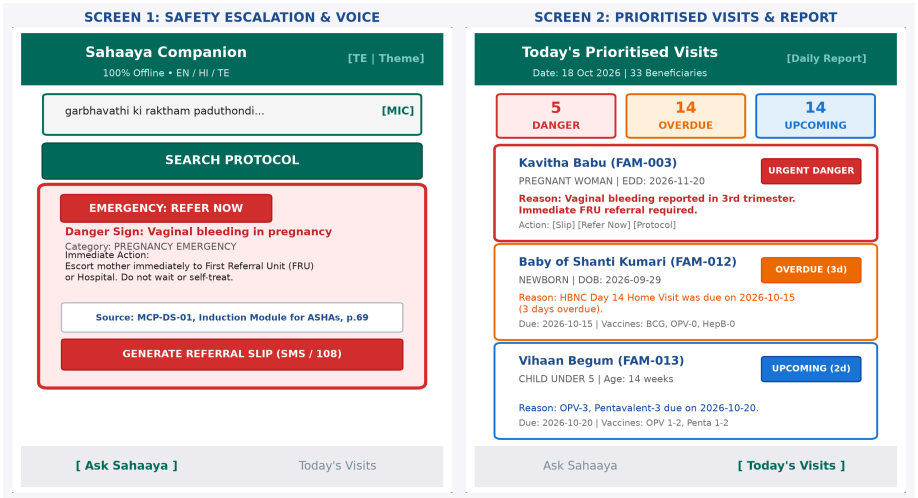
## Market Opportunity

- **Frontline Scale & National Need:**
  10.29 lakh ASHAs and ~2.5 lakh Sub-Centre ANMs across India serving 1,000 population norms, requiring an intuitive, reliable paramedical companion.
- **Critical Gap in Existing Systems:**
  State tools (ANMOL, mSakhi, Kilkari) suffer from cloud latency, sync failures, or lack deterministic emergency danger escalation and scope gating.
- **Sahaaya High-Impact Value:**
  100% offline, protocol-grounded companion: deterministic danger-sign escalation, doctor redirects, prioritized visits, emergency referral slips, and 2G SMS field sync.

# Proposed Solution

## Solution Overview

- **Deterministic Danger Escalation + Referral Slip:**
  Scans queries across English, Hinglish & Telugu-Latin; triggers red screen (100% recall on 367 phrasings) + 1-tap 108 Emergency Referral Slip.
- **Strict Non-Prescribing Scope Gate:**
  Hard-blocks antibiotic & dosage questions with physician redirects, ensuring paramedical workers operate strictly within ASHA guidelines.
- **Extractive BM25 Protocol Retrieval:**
  Pure Kotlin BM25 matches questions to official MoHFW/NHM manuals with exact document & page citations (0.53 ms latency, 0% hallucination).
- **Prioritised Daily Household Action Plan & 2G Sync:**
  Triage ranking: Danger Flags > Overdue Follow-ups > Routine Due Dates + 1-Tap End-of-Day PHC Summary Report (transmissible via 2G SMS).



# Design Steps & Architecture

## Technical Implementation

1. **Input & Offline Voice Recognition:**
   Supports speech and text queries in English, Hindi & Telugu (Latin). Trilingual instant switching and sunlight-readable dark/light modes.
2. **Layer 1: Deterministic Danger Rule Engine:**
   Hard-coded regex & phrase matching across 27 danger signs (100% recall). Immediately generates Red Alert + 108 Emergency Referral Slip.
3. **Layer 2: Intent Safety Gate:**
   Blocks clinical prescription/antibiotic queries with fixed doctor redirect, protecting workers against medical malpractice.
4. **Layer 3: Pure Kotlin BM25 Retriever:**
   Tokenises with code-mixed synonym expansion, scores against grounded MoHFW chunks with confidence thresholding (0% hallucination).
5. **Dynamic Visit-List Engine & 2G Sync:**
   Ranks 30+ family records by clinical urgency: Danger Flags -> Overdue -> Upcoming. 1-Tap generates daily PHC SMS field summaries.

## Architecture Diagram

# Challenges & Mitigation

## Challenge 1

**Problem:** Zero mobile reception in rural hamlets and severe RAM/battery constraints on low-end ASHA smartphones.

**Mitigation:** 100% on-device architecture with ZERO network permissions. Measured APK size is 16.9 MB, median response latency is 0.53 ms, and peak heap RAM footprint is < 1 MB.

## Challenge 2

**Problem:** Medical hallucination risks and catastrophic delays from missed pregnancy and newborn danger signs.

**Mitigation:** Deterministic danger rules execute BEFORE any retrieval logic (achieving 100% recall on 367 verified multilingual phrasings). Answers strictly cite section and page.

## Challenge 3

**Problem:** Code-mixed vernacular queries (English, Hinglish, Telugu in Latin script) with spelling variations and typos.

**Mitigation:** Engineered robust phonetic tokenisation, code-mixed synonym mapping, offline voice mic, and 1-tap referral/daily sync. On-device SLM planned for finale using organizer models.

# Unique Selling Proposition (USP)

## Key Differentiator 1

Deterministic Safety-First Layer:
Hard-coded emergency rule engine fires BEFORE any AI logic, achieving 100% recall (367/367 tested phrasings) with 0% hallucination + 1-tap 108 Emergency Referral Slip generation.

## Key Differentiator 2

100% Offline & Measured Speed:
Zero network permissions required.
Measured on JVM & physical phone:
· Median Latency: 0.53 ms
· P95 Latency: 5.38 ms
· APK Size: 16.9 MB | RAM: < 1 MB
Runs natively in Airplane Mode.

## Key Differentiator 3

Prioritised Household Action Plan:
Converts static paper registers into an automated daily workflow:
1. URGENT: Danger Flag Cases
2. OVERDUE: Ranked by days overdue
3. UPCOMING: Nearest scheduled care
Includes 1-Tap 2G SMS Daily Sync.

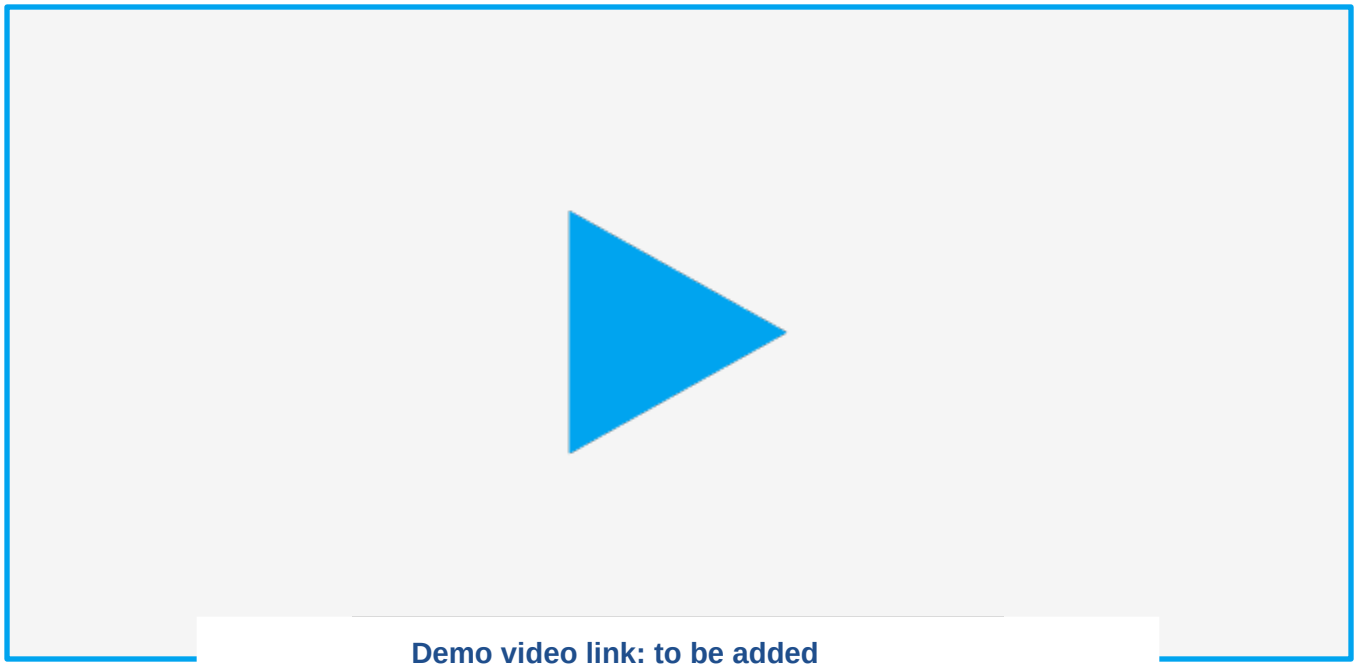
## Competitive Advantage Summary

· **Clinically Grounded Trust:**
Unlike generic LLM chatbots that hallucinate dosages, Sahaaya hard-refuses prescriptions, cites official MoHFW protocols, and produces actionable 108 ambulance referral slips.

· **Workflow Unification:**
Replaces fragmented flipbooks, complex checklists, and multi-app fatigue with a clean two-screen trilingual interface (EN/HI/TE) with offline voice input tailored for village health workers.

# Demo & POC Video

## Live Demonstration



**Demo video link: to be added**

Working Offline Android POC (app-debug.apk, 16.9 MB) | 100% Danger Recall (367 phrasings) | 0.53 ms Median Latency | Zero Cloud Permissions